

ENCLOSURE TO
FLEET MONTHLY
ROR – MAY 2015

FLEET RULE OF THE ROAD TEST FOR MAY 2015

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E) Vol I and the Mariners Handbook (NP100).

BR454 "A Guide to the Collision Avoidance Rules" 6th Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. A newer 7th Edition is under review by FLEET Nav. Until a decision is made on whether to issue the new edition no further copies will be supplied. Care should be taken when utilising the 6th Edition as the information contained within may not reflect the most recent changes to the regulations.

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff. The exam should take no longer than 1 hr.

In this test, any change in the status of the ship or of the environment will be indicated in the script. Part 2, for which there are no marks is only intended to increase awareness with reference to the appropriate publications and should not detract from the written exam.

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

Answers are provided for uniformity of marking. This test follows the same format as the Ship Command Rule of the Road paper.

You are the OOW on HMS ST ALBANS (length 133 metres), underway in the South Coast Exercise Areas conducting preparations for OST. Weather is fair and visibility is good, wind SW 12-15 kts and is forecast to increase. There are several serials planned throughout the day.

1. There is a large amount of adjacent area activity as well as numerous fishing vessels within your own areas. How should you maintain a proper look-out? (R. 5) **(2)**

As you proceed south, you sight several merchant vessels transiting through the areas at a range of 7 nm from your unit. You begin to assess them prior to making shipping reports to the CO.

2. a. In determining if risk of collision exists, what considerations should be taken into account? (R. 7d) **(2)**

- b. What should assumptions not be based on? (R. 7c) **(1)**

All the merchant vessels pass well clear.

3. a. You sight a large sailing vessel at green 40 at a range of 3 nm. What are your obligations with respect to this vessel? (R. 18) **(1)**
- b. Would these obligations change if the sailing vessel was overtaking your unit within the meaning of the definition? Fully explain your answer. (R. 18 & 13) **(3)**

Your SOA to the next RV is 21 kts. You start a gas turbine and increase speed.

4. In determining a safe speed, what factors shall be among those taken into account by all vessels? (R. 6a) **(3)**

5. On transit, you sight a power-driven vessel at green 60 at a range of 4 nm. You are 20° on her port bow and are the give-way vessel in a crossing situation.

- a. What, so far as possible, is the give-way vessel directed to do? (R. 16) **(1)**

- b. What is the stand-on vessel directed to do? (R. 17) **(1)**

- c. When may the stand-on vessel take action? (R. 17) **(2)**

6. You report the above contact to the CO and recommend an alteration of course.

- a. What is the direction on taking action to avoid collision? (R. 8a) **(2)**

- b. If there is sufficient sea room, what may be the most effective action to avoid a close-quarters situation? (R. 8c) **(2)**

You alter to starboard and the power-driven vessel passes clear.

7. You sight a vessel indicating she is constrained by her draught. What is the definition of a vessel constrained by draught? (R. 3h) **(2)**

8. Continuing on transit to your RV, you sight units conducting replenishment at sea. The units are indicating they are restricted in ability to manoeuvre.

- a. What is the definition of a vessel restricted in her ability to manoeuvre? (R. 3g) **(1)**

- b. Excluding vessels engaged in replenishment operations, what other classes of vessel does the term 'restricted in her ability to manoeuvre' include? (R. 3g) **(3)**

- c. What is the caveat stated on this list of vessels? (R. 3g) **(1)**

9. You note that the replenishment group is approaching a unit engaged in the launching and recovery of aircraft. Which unit is directed to keep out of the way of the other? (ATP1E) **(1)**

10. On passing the replenishment group safely, you close your RV to find a tug towing a barge underway through your allocated area.

- a. The tug is not showing shapes that would indicate she is restricted in ability to manoeuvre. Should she be? Justify your answer. (R. 3g) **(3)**

b. How would the towing vessel and tow indicate the fact if the length of the tow exceeded 200m? (R. 24) **(2)**

c. How is the length of tow measured? (R. 24a) **(1)**

The tow clears your areas just as visibility begins to decrease.

11. What is the definition of restricted visibility? (R. 3I) **(3)**

You assess visibility to be 2.5 nm.

12. When does Rule 19 apply? (R. 19a) **(3)**

Visibility continues to decrease to 1.5 nm.

13. You detect by the radar alone the presence of another vessel. What should you do? (R. 19d) **(4)**

14. Detail your actions on hearing apparently forward of the beam the fog signal of another vessel. (R. 19e) **(2)**

15. Should you be sounding sound signals? If so, what should they be? (R. 35) **(2)**

The visibility begins to improve and you continue on transit.

16. You arrive at your RV position. When do the rules in Part B Section II apply over those listed in Section III? (R. 11) **(1)**

You arrive at your RV with visibility at 6 nm; wind has increased to 20 kts. You stop in the water to commence preparations for a propulsion check.

17. You sight a vessel at G45 at 4 nm. You are 10° on his port bow. Are you still give-way in a crossing situation if stopped in the water? (R. 15) **(2)**

18. You commence the propulsion check, during your serial you sight units formed in a screen conducting an ADEX. The screen clearly consists of a main body and accompanying screen units.

a. What is the responsibility of screen units with respect to the main body? (ATP 1E) **(1)**

b. Does Rule 15 of IRPCS apply between screen ships and ships of the main body? (ATP 1E) **(2)**

Propulsion check complete, you note that sunset is approaching.

19. The Rules concerning lights shall be complied with from sunset to sunrise, and during such times no other lights shall be exhibited, except... what? (R. 20b) **(3)**

20. List the specific characteristics and arc of visibility of your masthead lights. (R. 21a) **(3)**

After a weekend in Plymouth the Ship is alongside HMNB Devonport, 5W, bows N and is preparing to sail for further OST preparations. Weather is fair, visibility is 6 nm and wind is Westerly at 10-15 kts. You are sailing 1 hour after high water and 1 hour prior to sunrise. You are SSD OOW for the departure.

The CO manoeuvres the ship off the berth and the NO takes the ship to conduct the outbound pilotage. You begin assessing shipping in the Hamoaze as you start to head south.

21. You sight the navigation lights of a small power driven vessel crossing the Hamoaze from Torpoint. She is 40° on your starboard bow and her bearing is drawing very slowly left.

a. What is she directed to do? (R. 9b) **(2)**

- b. The vessel continues to draw slowly left with a CPA assessed as within 50 yards ahead. What action do you recommend to the NO? Justify your answer. (R. 9d, 8fiii, 34d) **(4)**

You take the requisite action and clear the vessel above.

22. You sight a sailing vessel under power transiting parallel to the main navigation channel. There is no one visible at the wheel in the early morning light. Nothing in these Rules shall exonerate any vessel, the owner, master or crew thereof, from what? (R. 2a) **(2)**

You clear the sailing vessel and enter the narrows.

23. You sight a vessel in the Sound displaying two all-round lights in a vertical line, the upper being white and the lower red. You can also see a sidelight. What is this vessel? (R. 29) **(2)**

24. On manoeuvring into the Sound, you see a vessel displaying an all-round white light forward and aft, the aft light being lower than the light forward. She has also illuminated her decks. What can you state about this vessel? (R. 30a, c) **(2)**

You pass Penlee Point and indicate sunrise to the CO; visibility remains at 4 nm as you begin to head south.

25. To port, you see HMS BULWARK docked down and engaged in the launching and recovery of landing craft utilities. She is indicating she is restricted in ability to manoeuvre.

- a. What shapes should she be displaying? (R. 27bii) **(1)**
- b. While conducting her current operations, would HMS BULWARK have priority over a vessel engaged in the launching and recovery of aircraft? (ATP 1E) **(1)**

26. Just visible ahead at 4 nm is a merchant vessel. You assess you are 170° on her starboard bow.

- a. What situation is this and why? (R.13b) **(2)**

b. What is your obligation with respect to the merchant vessel? (R.13a) **(1)**

27. You manoeuvre clear of the merchant vessel but soon sight another, much larger vessel 50° on your port bow. You are on her starboard bow at a range of 5 nm.

a. You assess the vessel is potentially VLCC size. Does this size impact on her responsibilities in accordance with IRPCS? **(1)**

b. Who is the give-way vessel? (R. 15) **(1)**

28. Passing the large vessel safely, you sight a vessel still displaying navigation lights despite the fact sunrise has now passed and visibility is opening to 10 nm. She is indicating she is trawling.

a. What is the definition of a vessel engaged in trawling? (R. 26b) **(1)**

b. What lights should she be showing if trawling, underway and over 50m? (R. 26b) **(1)**

29. What is the definition of a vessel engaged in fishing? (R. 3d) **(2)**

30. What sound signal would a vessel engaged in fishing sound in restricted visibility? (R. 35c) **(2)**

Total _____ /85

ANSWERS

You are the OOW on HMS ST ALBANS (length 133 metres), underway in the South Coast Exercise Areas conducting preparations for OST. Weather is fair and visibility is good, wind SW 12-15 kts and is forecast to increase. There are several serials planned throughout the day.

1. There is a large amount of adjacent area activity as well as numerous fishing vessels within your own areas. How should you maintain a proper look-out? (R. 5) **(2)**

Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate to the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.

As you proceed south, you sight several merchant vessels transiting through the areas at a range of 7 nm from your unit. You begin to assess them prior to making shipping reports to the CO.

2. a. In determining if risk of collision exists, what considerations should be taken into account? (R. 7d) **(2)**

(i) **Such risk shall be deemed to exist if the compass bearing of and approaching vessel does not appreciably change;**

(ii) **Such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range.**

- b. What should assumptions not be based on? (R. 7c) **(1)**

Assumptions should not be made on the basis of scanty information, especially scanty radar information.

All the merchant vessels pass well clear.

3. a. You sight a large sailing vessel at green 40 at a range of 3nm. What are your obligations with respect to this vessel? (R. 18) **(1)**

A power-driven vessel shall keep out of the way of a sailing vessel.

- b. Would these obligations change if the sailing vessel was overtaking your unit within the meaning of the definition? Fully explain your answer. (R. 18) **(3)**

Yes, except where Rule 13 otherwise requires, a power-driven vessel shall keep out of the way of a sailing vessel. Therefore, in this case, Rule 13 takes precedence and the sailing vessel is required to keep out your way.

Your SOA to the next RV is 21 kts. You start a gas turbine and increase speed.

4. In determining a safe speed, what factors shall be among those taken into account by all vessels? (R. 6a) **(3)**

(i) **The state of visibility;**

(ii) **the traffic density including concentrations of fishing vessels or any other vessels;**

- (iii) **The manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;**
- (iv) **at night the presence of background lights such as from shore lights or from back scatter of her own lights;**
- (v) **the state of wind, sea and current, and the proximity of navigational hazards;**
- (vi) **the draught in relation to the available depth of water.**

5. On transit, you sight a power-driven vessel at green 60 at a range of 4nm. You are 20° on her port bow and are the give-way vessel in a crossing situation.

- a. What, so far as possible, is the give-way vessel directed to do? (R. 16) **(1)**

Take early and substantial action to keep well clear.

- b. What is the stand-on vessel directed to do? (R. 17) **(1)**

Keep her course and speed.

- c. When may the stand-on vessel take action? (R. 17) **(2)**

The stand-on vessel may take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these Rules.

6. You report the above contact to the CO and recommend an alteration of course.

- a. What is the direction on taking action to avoid collision? (R. 8a) **(2)**

Any action to avoid collision shall be taken in accordance with the rules of this part and, if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship.

- b. If there is sufficient sea room, what may be the most effective action to avoid a close-quarters situation? (R. 8c) **(2)**

Alteration of course alone may be the most effective action to avoid a close quarter situation provided that it is made on good time, is substantial and does not results in another close-quarters situation.

You alter to starboard and the power-driven vessel passes clear.

7. You sight a vessel indicating she is constrained by her draught. What is the definition of a vessel constrained by draught? (R. 3h) **(2)**

A power-driven vessel which because of her draught in relation to the available depth and width of navigable water is severely restricted in her ability to deviate from the course she is following.

8. Continuing on transit to your RV, you sight units conducting replenishment at sea. The units are indicating they are restricted in ability to manoeuvre.

- a. What is the definition of a vessel restricted in her ability to manoeuvre? (R. 3g) **(1)**

A vessel which from the nature of her work is restricted in her ability to manoeuvre as required by these Rules and is therefore unable to keep out of the way of another vessel.

b. Excluding vessels engaged in replenishment operations, what other classes of vessel does the term 'restricted in her ability to manoeuvre' include? (R. 3g) (3)

(i) A vessel engaged in laying, servicing or picking up a navigation mark, submarine cable or pipeline;

(ii) a vessel engaged in dredging, surveying or underwater operations;

(iv) a vessel engaged in the launching or recovery of aircraft;

(v) a vessel engaged in mine clearance operations;

(vi) a vessel engaged in a towing operation such as severely restricts the towing vessel and her tow in their ability to deviate from their course.

c. What is the caveat stated on this list of vessels? (R. 3g) (1)

This includes vessels restricted in ability to manoeuvre but the definition is not limited to this list.

9. You note that the replenishment group is approaching a unit engaged in the launching and recovery of aircraft. Which unit has right of way? (ATP1E) (1)

Ships engaged in replenishment have right of way over carriers and other ships engaged in flight operations.

10. On passing the replenishment group safely, you arrive at your RV to find a tug towing a barge.

a. The tug is not showing shapes that would indicate she is restricted in ability to manoeuvre. Should she be? Justify your answer. (R. 3g) (3)

Shapes indicating that she is restricted in ability to manoeuvre are only required if she is engaged in a towing operation such as severely restricts the towing vessel and her tow in their ability to deviate from their course.

b. How would the towing vessel and tow indicate the fact if the length of the tow exceeded 200m? (R. 24a & g) (2)

When the length of the tow exceeds 200m, both the towing vessel and the vessel towed must exhibit a black diamond shape.

c. How is the length of tow measured? (R. 24a) (1)

From the stern of the towing vessel to the after end of the tow.

The tow clears your areas just as visibility begins to decrease.

11. What is the definition of restricted visibility? (R. 3l) (3)

The term 'restricted visibility' means any condition in which visibility is restricted by fog, mist, falling snow. Heavy rainstorms, sandstorms or any other similar causes.

You assess visibility to be 2.5 nm.

12. When does Rule 19 apply? (R. 19a) (3)

Rule 19 applies to vessels not in sight of one another when operating in or near an area of restricted visibility.

Visibility continues to decrease to 1.5 nm.

13. You detect by the radar alone the presence of another vessel. What should you do? (R. 19d) (4)

A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:

(i) An alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;

(ii) An alteration of course towards a vessel abeam or abaft the beam.

14. Detail your actions on hearing apparently forward of the beam the fog signal of another vessel. (R. 19e) (2)

A vessel which hears apparently forward of the beam the fog signal of another vessel shall reduce her speed to the minimum at which she can be kept on her course. She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over.

15. Should you be sounding sound signals? If so, what should they be? (R. 35) (2)

A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.

The visibility begins to improve and you continue on transit.

16. You arrive at your RV position. When do the rules in Part B Section II apply over those listed in Section III? (R. 11) (1)

Rules in this Section apply to vessels in sight of one another.

You arrive at your RV with visibility at 6 nm; wind has increased to 20 kts. You stop in the water to commence preparations for a propulsion check.

17. You sight a vessel at G45 at 4 nm. You are 10° on his port bow. Are you still give-way in a crossing situation if stopped in the water? (R. 15) (2)

Yes, a power driven vessel stopped must, unless she is not under command, or restricted in her ability to manoeuvre, keep out of the way of another vessel which approaches so as to involve risk of collision from any direction between right ahead and 22.5° abaft the beam on her starboard side.

18. You commence the propulsion check, during your serial you sight units formed in a screen conducting an ADEX. The screen clearly consists of a main body and accompanying screen units.

a. What is the responsibility of screen units with respect to the main body? (ATP 1E) (1)

Ships of the screen are to keep clear of those of the main body.

- b. Does Rule 15 of IRPCS apply between screen ships and ships of the main body? (ATP 1E) (2)

Rule 15 of IRPCS is not to apply between screen ships and ships of the main body.

Propulsion check complete, you note that sunset is approaching.

19. The Rules concerning lights shall be complied with from sunset to sunrise, and during such times no other lights shall be exhibited, except... what? (R. 20b) (3)

Except such light as cannot be mistaken for the lights specified in these Rules or do not impair their visibility or distinctive character, or interfere with the keeping of a proper lookout.

20. List the specific characteristics and arc of visibility of the masthead light. (R. 21a) (3)

Masthead light means a white light placed over the fore and aft centreline of the vessel showing an unbroken light over an arc of the horizon of 225 degrees and so fixed as to show the light from right ahead to 22.5 degrees abaft the beam on either side of the vessel.

After a weekend in Plymouth, the Ship is alongside HMNB Devonport, 5W, bows N and is preparing to sail for further DCT preparations. Weather is fair, visibility is 5 nm and wind is Westerly 10-15 kts. You are sailing 1 hour after high water and 1 hour prior to sunrise. You are SSD OOW for the departure.

The CO manoeuvres the ship off the berth and the NO takes the ship to conduct the outbound pilotage. You begin assessing shipping in the Hamoaze as you start to head south.

21. You sight the navigation lights of a small power driven vessel crossing the Hamoaze from Torpoint. She is 40° on your starboard bow and her bearing is drawing very slowly left.

- b. What is she directed to do? (R. 9b, 9d) (2)

A vessel of less than 20 metres in length or a sailing vessel shall not impede the passage of a vessel which can navigate safely only within a narrow channel or fairway.

A vessel shall not cross a narrow channel or fairway if such crossing impedes the passage of a vessel which can navigate safely only within such narrow channel or fairway. The latter vessel may use the sound signal prescribed in Rule 34(d) if in doubt as to the intention of the crossing vessel.

- b. The vessel continues to draw slowly left with a CPA assessed as within 50 yards ahead. What action do you recommend to the NO? Justify your answer. (R. 9d, 8fiii, 15, 34d) (4)

Sound 5 short blasts in accordance with Rule 9d. Reduce speed and stop if necessary (there is not sufficient sea room to alter course). A vessel the passage of which is not to be impeded remains fully obliged to comply with the rules of this part when two vessels are approaching one another so as to involve risk of collision.

You take the requisite action and clear the vessel above.

22. You sight a sailing vessel under power transiting parallel to the main navigation channel. There is no one visible at the wheel in the early morning light. Nothing in these Rules shall exonerate any vessel, the owner, master or crew thereof, from what? (R. 2a) **(2)**

The consequences of any neglect to comply with these Rules or the neglect of any precaution which may be required by the ordinary practice of seaman or by the special circumstances of the case.

You clear the sailing vessel and enter the narrows.

23. You sight a vessel in the Sound displaying two all-round lights in a vertical line, the upper being white and the lower red. You can also see a sidelight. What is this vessel? (R. 29) **(2)**

A vessel engaged in pilotage duty, underway.

24. On manoeuvring into the Sound, you see a vessel displaying an all-round white light forward and aft, the aft light being lower than the light forward. She has also illuminated her decks. What can you state about this vessel? (R. 30a, c) **(2)**

A vessel at anchor, >100 metres.

You pass Penlee Point and indicate sunrise to the CO; visibility remains at 4nm as you begin to head south.

25. To port, you see HMS BULWARK docked down and engaged in the launching and recovery of landing craft utilities. She is indicating she is restricted in ability to manoeuvre.

b. What shapes should she be displaying? (R. 27bii) **(1)**

Three shapes in a line where they can best be seen. The highest and lowest of these shapes shall be balls and the middle one a diamond.

b. While conducting her current operations, would HMS BULWARK have priority over a vessel engaged in the launching and recovery of aircraft? (ATP 1E) **(1)**

Yes

26. Just visible ahead at 4 nm is a merchant vessel. You assess you are 170° on her starboard bow.

c. What situation is this and why? (R.13b) **(2)**

This is an overtaking situation. A vessel shall be deemed to be overtaking when coming up with another vessel from a direction more than 22.5° abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.

b. What is your obligation with respect to the merchant vessel? (R.13a) **(1)**

Any vessel overtaking any other shall keep out of the way of the vessel being overtaken.

27. You manoeuvre clear of the merchant vessel but soon sight another, much larger vessel 50° on your port bow. You are on her starboard bow at a range of 5 nm.

a. You assess the vessel is potentially VLCC size. Does this size impact on her responsibilities in accordance with IRPCS? **(1)**

No, assuming she is not indicating constrained by draught.

b. Who is the give-way vessel? (R. 15) **(1)**

The VLCC is give-way

28. Passing the large vessel safely, you sight a vessel still displaying navigation lights despite the fact sunrise has now passed and visibility is opening to 10 nm. She is indicating she is trawling.

a. What is the definition of a vessel engaged in trawling? (R. 26b) **(1)**

A vessel engaged in trawling, by which is meant the dragging through the water of a dredge net or other apparatus used as a fishing appliance.

b. What lights should she be showing if trawling, underway and over 50m? (R. 26b) **(2)**

Two all-round lights in a vertical line, the upper being green and the lower white, masthead abaft and higher than all-round green light, sidelights and sternlight.

29. What is the definition of a vessel engaged in fishing? (R. 3d) **(2)**

Means any vessel fishing with nets, lines, trawls or other fishing apparatus which restricts manoeuvrability, but does not include a vessel fishing with trolling lines or other fishing apparatus which do not restrict manoeuvrability.

30. What sound signal would a vessel engaged in fishing sound in restricted visibility? (R. 35c) **(2)**

Shall sound, at intervals of not more than 2 minutes three blasts in succession, namely one prolonged followed by two short.

Total /85